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(54) **TURBO-BOOST CONTROLLED INTAKE SYSTEM**

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F02M 35/10 (2006.01)

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USPC 123/383
See application file for complete search history.

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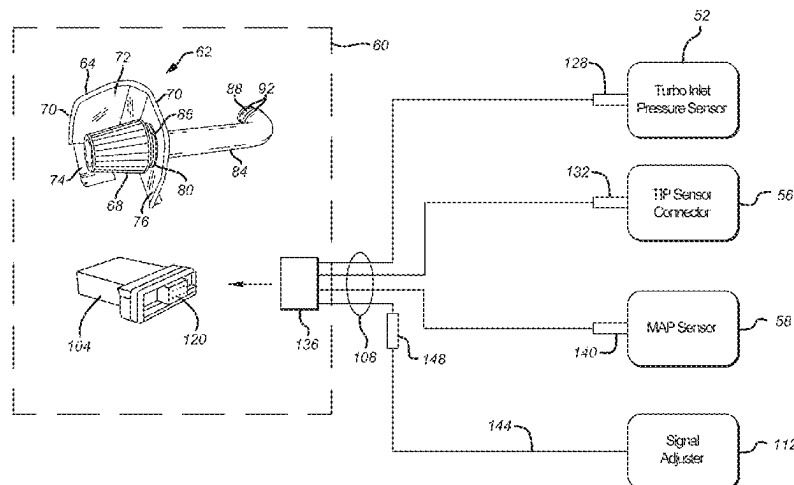
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(57)

ABSTRACT

A turbo-boost controlled intake system is disclosed that provides a driver of a vehicle with greater control over vehicle performance. The turbo-boost controlled intake system includes a control module that is coupled with an aircharger air intake. The control module instructs an electronic control unit of the vehicle to increase manifold pressure to a higher level before releasing the pressure through a waste gate so as to provide a greater power output of the engine. The turbo-boost controlled intake system further includes a wiring harness and a signal adjuster. The wiring harness couples the control module with a turbo inlet pressure sensor, a manifold absolute pressure sensor, and an electronic control unit of the vehicle. The signal adjuster includes a rheostat that enables manual adjustment of the power output of the engine.

18 Claims, 4 Drawing Sheets



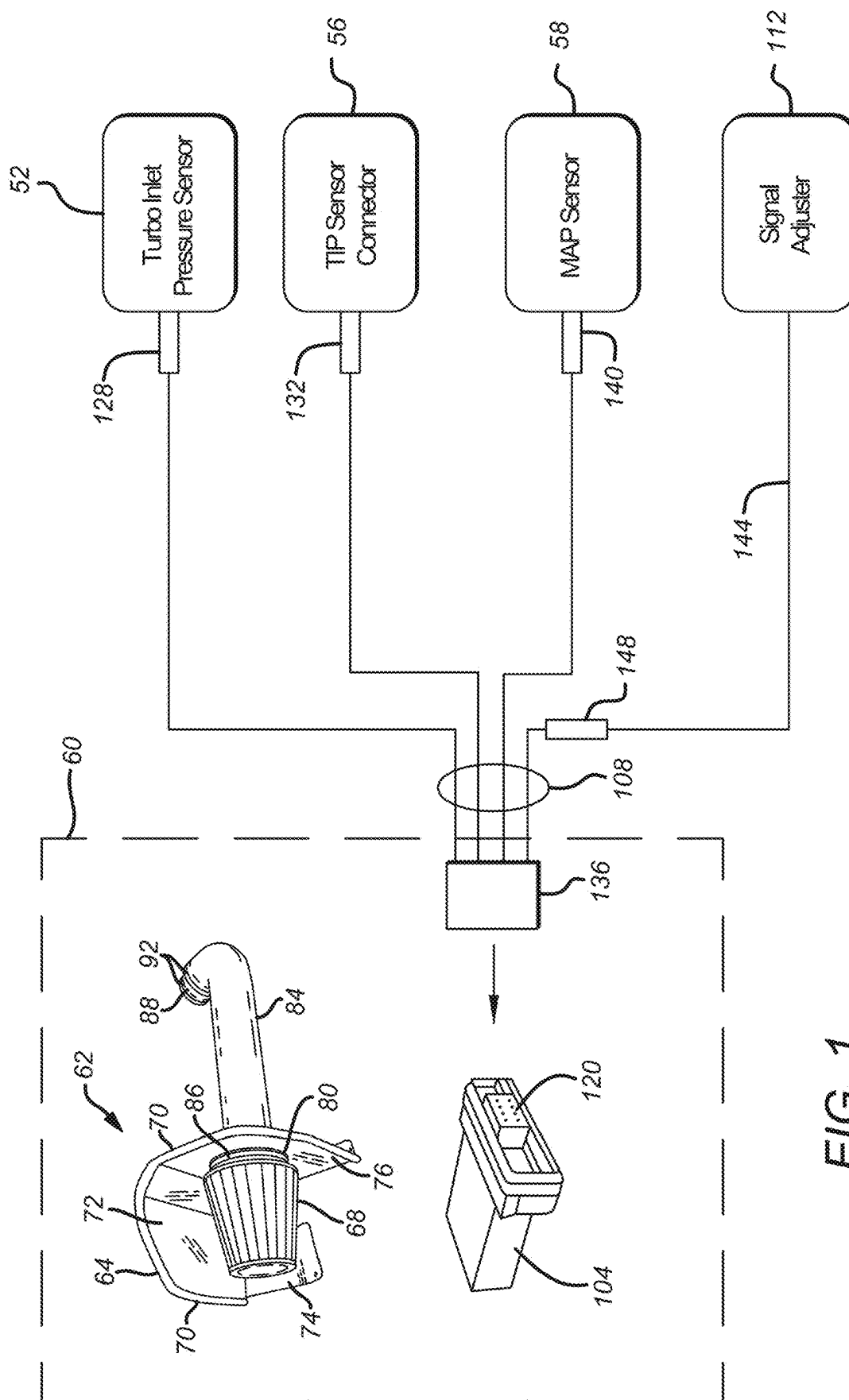
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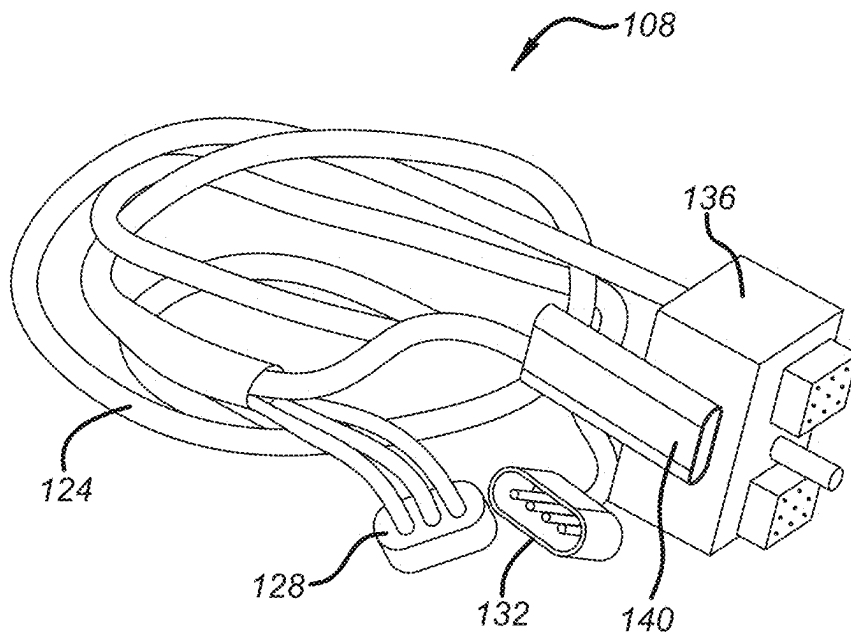
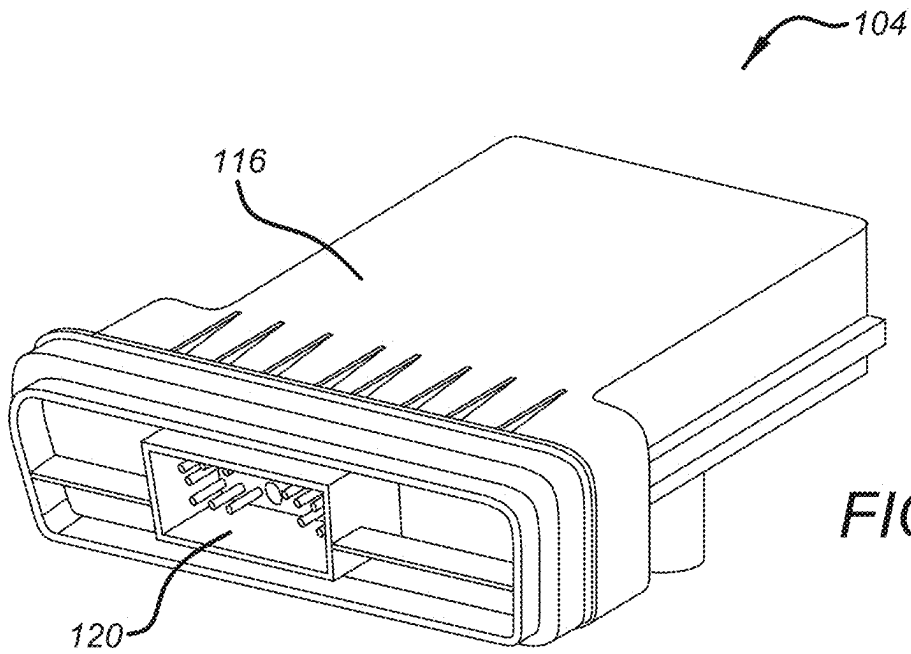


FIG. 4

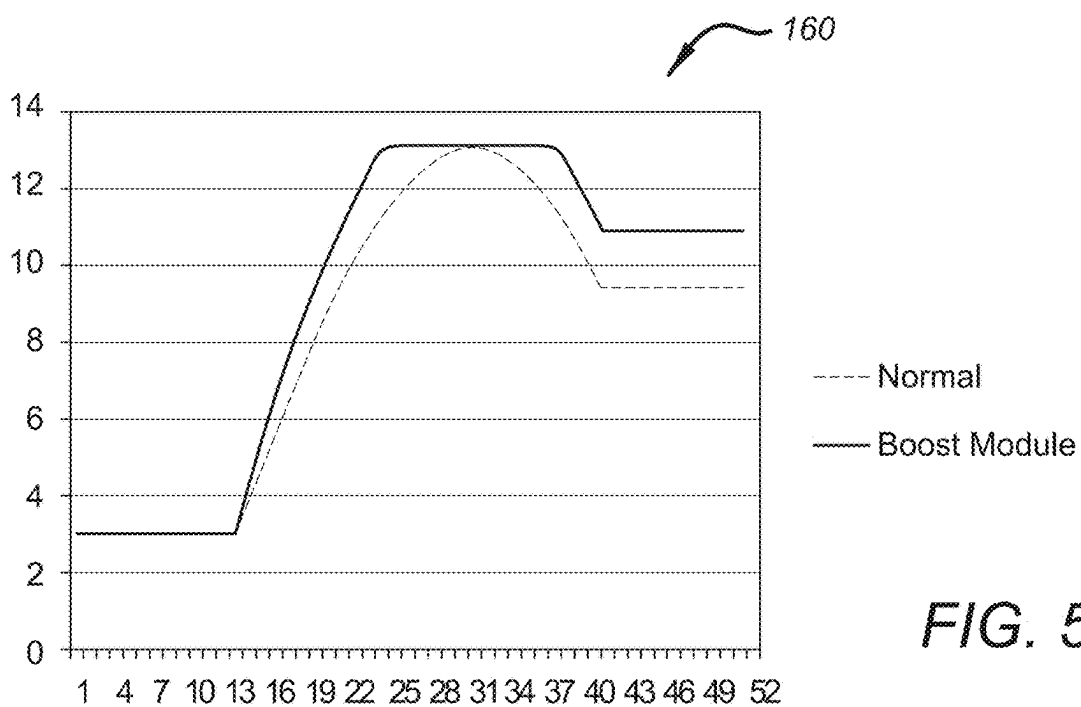
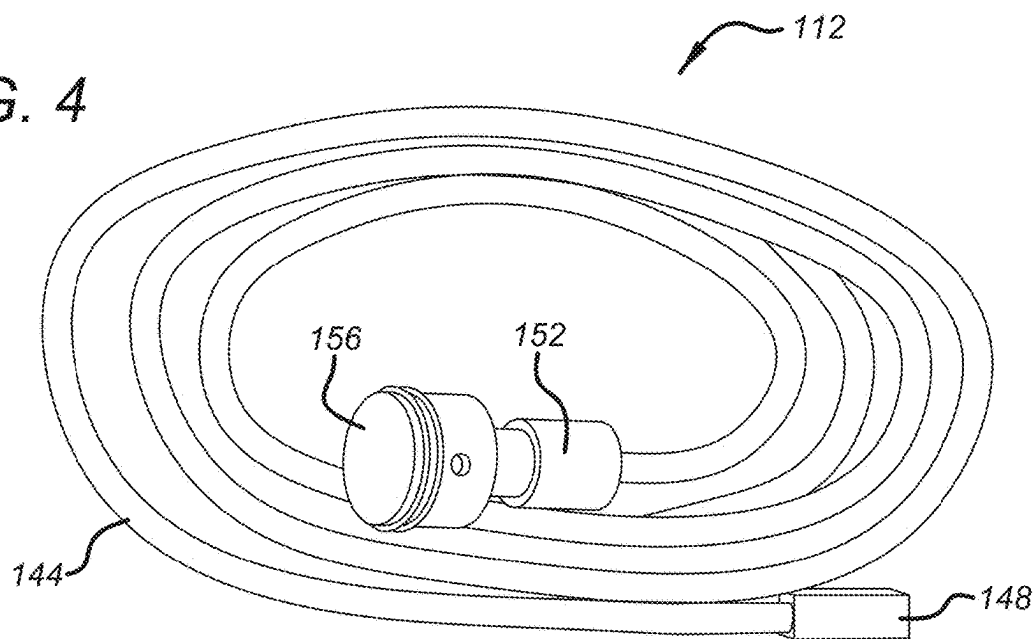


FIG. 5

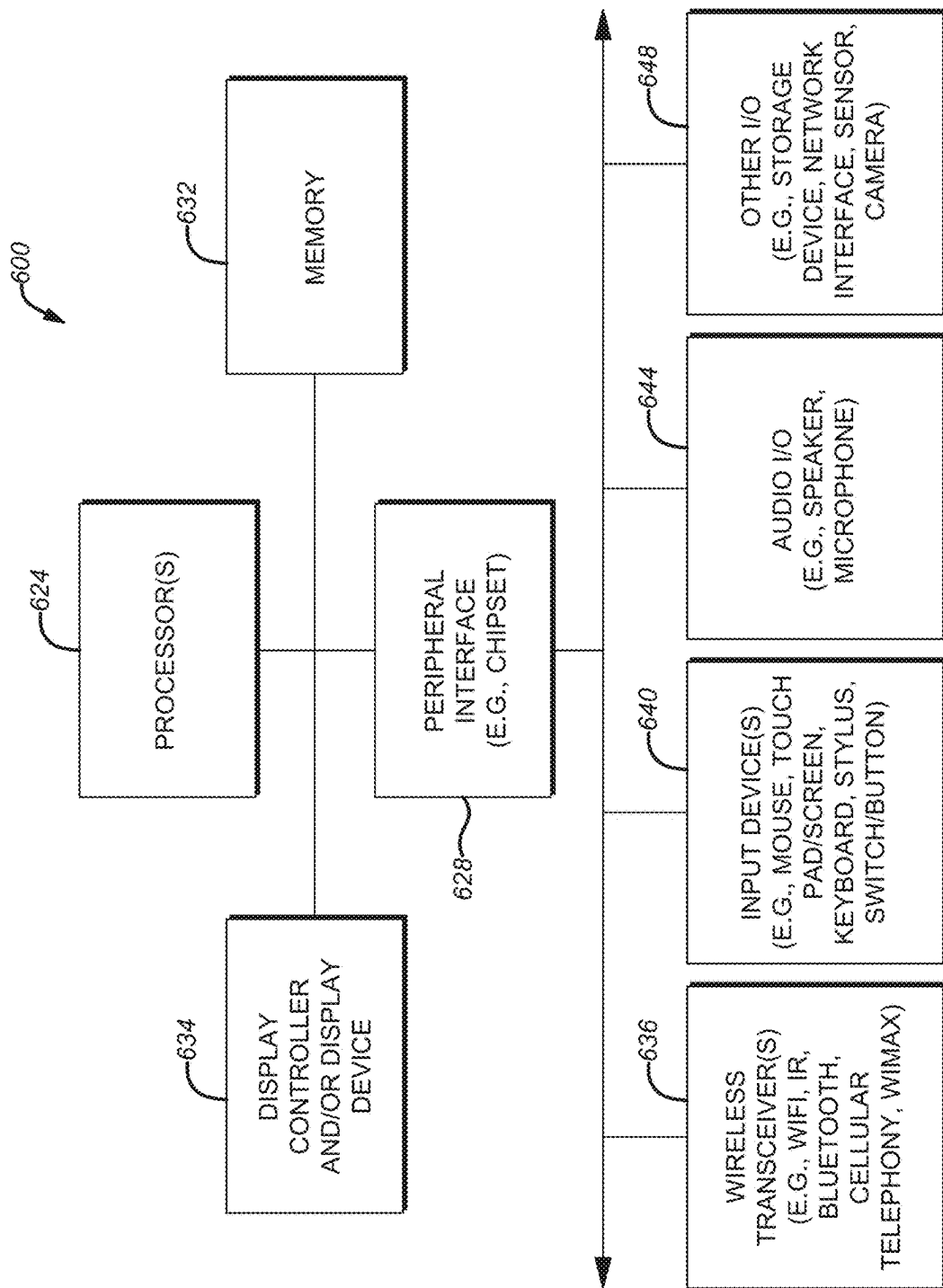


FIG. 6

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**TURBO-BOOST CONTROLLED INTAKE
SYSTEM****PRIORITY**

This application claims the benefit of and priority to U.S. patent application Ser. No. 18/090,078 filed Dec. 28, 2022 and U.S. patent application Ser. No. 16/937,326 filed on Jul. 23, 2020 and U.S. Provisional Application, entitled “Turbo-Boost Controlled Intake System,” filed on Jul. 24, 2019 and having application Ser. No. 62/878,256, the entirety of said application being incorporated herein by reference.

FIELD

Embodiments of the present disclosure generally relate to the field of vehicle control systems. More specifically, embodiments of the disclosure relate to a turbo-boost controlled intake system that provides greater control over the power output of turbocharged engines.

BACKGROUND

A turbocharger is generally a turbine-driven, forced induction device configured to increase the efficiency and power of an engine. As compared to a naturally aspirated engine, a turbocharged engine produces greater output power because the turbine forces more air, and proportionately more fuel, into the engine’s combustion chambers than atmospheric pressure alone. As will be appreciated by those skilled in the art, turbochargers were once referred to as “turbosuperchargers” when all forced induction devices were classified as “superchargers.” At present, however, the term “supercharger” typically is used in reference to only mechanically driven, forced induction devices, such as by way of a belt, gear, shaft, or chain connected to the engine’s crankshaft, whereas the term “turbocharger” is used in reference to a turbine driven by the engine’s exhaust gas. Turbochargers find wide use with truck, car, train, aircraft, and construction equipment engines. Turbochargers typically are used with Otto cycle and Diesel cycle internal combustion engines; although more recently, turbochargers have also been found to be useful with automotive fuel cells.

A drawback to many factory turbocharged engines is that, under factory parameters, a vehicle’s waste gate releases manifold pressure at a level specified by the vehicle manufacturer, thereby undesirably dropping the available power level of the vehicle. What is needed, therefore, is a turbo-boost control module capable of raising an amount of pressure within the engine’s manifold to a higher level before releasing it through the waste gate, thereby maintaining more boost for when it is desired.

SUMMARY

An apparatus and methods are provided for a turbo-boost controlled intake system that provides a driver of a vehicle with greater control over vehicle performance. The turbo-boost controlled intake system includes a control module that is coupled with an aircharger air intake. The control module is configured to instruct an electronic control unit of the vehicle to increase manifold pressure to a higher level before releasing the pressure through a waste gate so as to provide a greater power output of the engine. The turbo-boost controlled intake system may further include a wiring harness and a signal adjuster. The wiring harness is configured to couple the control module with a turbo inlet pressure

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sensor, a manifold absolute pressure sensor, and an electronic control unit of the vehicle. The signal adjuster includes a rheostat that is configured to enable manual adjustment of the power output of the engine.

5 In an exemplary embodiment, a turbo-boost controlled intake system configured for providing greater control over vehicle performance comprises: an aircharger air intake system for conducting a filtered airstream to an air intake of an engine; a control module coupled with the aircharger air intake system and configured to signal an increase in manifold pressure before releasing the pressure through a waste gate so as to maintain additional boost for an increased power output of the engine; and a wiring harness configured to electrically couple the control module with a turbo inlet pressure sensor, a manifold absolute pressure sensor, and an electronic control unit of the vehicle.

In another exemplary embodiment, the aircharger air intake system comprises: an air filter comprising a filter medium configured to entrap particulates flowing within an airstream; an air box including one or more sidewalls and a mount wall; an intake tube coupled with the air filter and configured to conduct the airstream to the air intake; and an adapter configured to couple the intake tube with the air intake. In another exemplary embodiment, the one or more sidewalls are configured to receive fasteners suitable for installing the air box onto the engine. In another exemplary embodiment, the mount wall comprises an opening that receives an adapter configured to couple the air filter and the intake tube to the mount wall. In another exemplary embodiment, the intake tube comprises a shape and size suitable for conducting the airstream from the air filter into the air intake.

In another exemplary embodiment, the control module is comprised of one or more microprocessors that can process input signals received from the turbo inlet pressure sensor and the manifold absolute pressure sensor. In another exemplary embodiment, the control module includes an internal lookup table whereby turbo inlet pressure sensor and manifold absolute pressure sensor readings may be evaluated. In another exemplary embodiment, the control module includes an input socket configured to receive a signal connector comprising the wiring harness. In another exemplary embodiment, the input socket couples the control module with turbo inlet pressure sensor, the manifold absolute pressure sensor, and the electronic control unit of the vehicle.

In another exemplary embodiment, the wiring harness includes a cable, a turbo inlet pressure sensor connector, a turbo inlet pressure sensor harness connector, a signal connector, and a manifold absolute pressure sensor connector. In another exemplary embodiment, the turbo inlet pressure sensor connector is configured to be coupled directly with the turbo inlet pressure sensor of the vehicle, and wherein the turbo inlet pressure sensor harness connector is configured to be coupled with the wiring harness that was originally coupled with the turbo inlet pressure sensor. In another exemplary embodiment, the signal connector is configured to be plugged into an input socket of the control module. In another exemplary embodiment, the manifold absolute pressure sensor connector is configured to be coupled with the manifold absolute pressure sensor of the vehicle for the purpose reading air pressure within the engine manifold.

In another exemplary embodiment, the turbo-boost controlled intake system further includes a signal adjuster configured to enable manual adjustment of the power output of the engine. In another exemplary embodiment, the signal adjuster comprises a cable that extends from a controller

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connector to a rheostat. In another exemplary embodiment, the controller connector is configured to be plugged into a controller socket comprising the wiring harness. In another exemplary embodiment, the signal adjuster includes a control dial configured to be coupled with the rheostat to facilitate hand operation of the rheostat.

In an exemplary embodiment, a method for a turbo-boost controlled intake system to provide greater control over engine performance of a vehicle comprises: configuring an aircharger air intake system for conducting a filtered airstream to an air intake of the engine; configuring a control module to signal an increase in manifold pressure before releasing the pressure through a waste gate so as to maintain additional boost for an increased power output of the engine; coupling the control module with the aircharger air intake system; and electrically coupling the control module turbo with an inlet pressure sensor, a manifold absolute pressure sensor, and an electronic control unit of the vehicle.

In another exemplary embodiment, electrically coupling includes configuring a wiring harness that includes a cable, a turbo inlet pressure sensor connector, a turbo inlet pressure sensor harness connector, a manifold absolute pressure sensor connector, and a signal connector capable of being plugged into an input socket of the control module. In another exemplary embodiment, configuring the control module includes configuring one or more microprocessors that can process input signals received from the turbo inlet pressure sensor and the manifold absolute pressure sensor. In another exemplary embodiment, configuring the aircharger air intake system comprises: providing an air filter comprising a filter medium to entrap particulates flowing within an airstream; configuring an air box to support the air filter and be coupled with the engine; coupling an intake tube between the air filter and the air intake of the engine; and adapting the air intake tube to conduct the airstream to the air intake of the engine.

BRIEF DESCRIPTION OF THE DRAWINGS

The drawings refer to embodiments of the present disclosure in which:

FIG. 1 illustrates an exemplary embodiment of a turbo-boost controlled intake system configured to provide a driver of a vehicle with greater control over vehicle performance;

FIG. 2 illustrates an exemplary embodiment of a control module configured to create an increase in manifold air pressure before releasing the pressure through a waste gate;

FIG. 3 illustrates an exemplary embodiment of a wiring harness configured to electrically couple the control module of FIG. 2 with a turbo inlet pressure sensor and a manifold absolute pressure sensor of the vehicle;

FIG. 4 illustrates an exemplary embodiment of a signal adjuster configured to facilitate manual adjustment of the turbo-boost of the vehicle;

FIG. 5 is a graph illustrating manifold pressure readings by an engine control unit of the vehicle with and without the turbo-boost control system of FIG. 1; and

FIG. 6 is a block diagram illustrating an exemplary data processing system that may be used with a turbo-boost control system according to the present disclosure.

While the present disclosure is subject to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and will herein be described in detail. The invention should be understood to not be limited to the particular forms disclosed, but on the contrary, the intention is to cover all

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modifications, equivalents, and alternatives falling within the spirit and scope of the present disclosure.

DETAILED DESCRIPTION

In the following description, numerous specific details are set forth in order to provide a thorough understanding of the present disclosure. It will be apparent, however, to one of ordinary skill in the art that the invention disclosed herein may be practiced without these specific details. In other instances, specific numeric references such as “first module,” may be made. However, the specific numeric reference should not be interpreted as a literal sequential order but rather interpreted that the “first module” is different than a “second module.” Thus, the specific details set forth are merely exemplary. The specific details may be varied from and still be contemplated to be within the spirit and scope of the present disclosure. The term “coupled” is defined as meaning connected either directly to the component or indirectly to the component through another component. Further, as used herein, the terms “about,” “approximately,” or “substantially” for any numerical values or ranges indicate a suitable dimensional tolerance that allows the part or collection of components to function for its intended purpose as described herein.

Factory turbocharged engines generally release manifold pressure at a level specified by the vehicle manufacturer, thereby undesirably dropping the available power level of the vehicle. The embodiments disclosed herein provide a turbo-boost control system capable of raising an amount of pressure within the engine's manifold to a higher level before releasing it through the waste gate, thereby maintaining more turbo-boost for greater power output of the engine.

FIG. 1 illustrates an exemplary embodiment of a turbo-boost controlled intake system 60 that is configured to provide a driver of a vehicle with greater control over vehicle performance. In the illustrated embodiment, the turbo-boost controlled intake system 60 includes an aircharger air intake 62 that is coupled with a control module 104, and a wiring harness 108. In some embodiments, a signal adjuster 112 may be included with the wiring harness 108. The turbo-boost controlled intake system 60 generally is configured to instruct an electronic control unit (ECU) of the vehicle to increase the manifold pressure to a higher level before releasing the pressure through a waste gate so as to provide a greater power output of the engine.

It is contemplated that the turbo-boost controlled intake system 60 is configured to provide a plug and play installation without requiring a practitioner to modify or fabricate components. As such, the control module 104 generally is integrated into the aircharger air intake 62, such that a practitioner need only install the aircharger air intake 62 and then connect the wiring harness 108 with the vehicle as described herein. The control module 104 may be integrated into the aircharger air intake 62 in a wide variety of different configurations without limitation, and without deviating beyond the scope of the present disclosure.

The aircharger air intake 62 generally comprises an air box 64 configured to couple an air filter 68 to an air intake of an engine. The air box 64 comprises sidewalls 72, a floor 74, and a mount wall 76. Preferably, the air box 64 is comprised of a rigid material that is sufficiently durable and temperature resistant to retain its configuration during installation and operation when coupled with the air intake of the engine. A pliable strip 70 is configured to be received along

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at least the top edges of the sidewalls **72** and the mount wall **76** so as to hide any sharp edges thereon.

The air filter **68** comprises a filter medium that provides an area to pass an airstream and entrap particulates and other contaminants flowing with the airstream. The filter medium may be comprised of paper, foam, cotton, spun fiberglass, or other known filter materials, woven or non-woven material, synthetic or natural, or any combination thereof. The filter medium may be pleated, or otherwise shaped, or contoured so as to increase a surface area for passing the airstream to be cleaned. The length of the filter medium in the circumferential direction may be longer than the circular circumference of the air filter **68** generally, such that the surface area of the filter medium is greater than the profile surface area of the air filter **68**. Further, the air filter **68** is not limited to the exact shape illustrated, but rather may include a wide variety of generally cylindrical shapes, generally circular, oval, round, curved, conical, or other closed perimeter shape, that provide a relatively large surface area in a given volume of the filter.

In some embodiments, the filter medium comprises 4 to 6 layers of cotton gauze sandwiched between two epoxy-coated aluminum wire screens. The cotton may be advantageously treated with a suitably formulated filter oil composition that causes tackiness throughout microscopic strands comprising the filter medium. The nature of the cotton allows high volumes of airflow, and when combined with the tackiness of the filter oil composition creates a powerful filtering medium which ensures a high degree of air filtration. Further details about components comprising the air filter **68**, as well as details about the filter oil composition, are disclosed in U.S. patent application Ser. No. 14/181,678, entitled "Air Box With Integrated Filter Media," filed on Feb. 16, 2014, and U.S. patent application Ser. No. 14/701,163, entitled "Filter Oil Formulation," filed on Apr. 30, 2015, the entirety of each of which is incorporated herein by reference.

The air box **64** generally is of an open variety, rather than being an enclosed air box as is conventionally utilized with many vehicles. The sidewalls **72** and the floor **74** serve to protect the air filter **68** from road debris, as well as to isolate the air filter **68** from hoses and other components that may be present within an engine bay of the vehicle. It will be recognized by those skilled in the art that the open air box **64** improves airflow to the air filter **68**, and thus decreases air resistance to the air intake of the engine, thereby improving engine performance beyond that otherwise possible with an enclosed air box. The air box **64** generally is configured to be mounted, or fastened, onto the engine. As such, the sidewalls **72** are configured to receive suitable fasteners to facilitate installing the air box **64** onto the engine. It is contemplated that the fasteners generally may comprise any of suitably designed holes, brackets, molded shaped portions, protrusions, extensions, angled brackets, hardware fasteners, or other any similar device for supporting the air box **64** within the engine bay. It should be understood that the particular fasteners will vary according to the specific make and model of the vehicle with which the air box **64** is to be used.

The mount wall **76** generally is configured to support the air filter **68** and provide an interface between the air filter **68** and the air intake of the engine. An opening **80** in the mount wall **76** is configured to receive an intake tube **84** that is configured to couple the air filter **68** with the air intake of the engine. In the embodiment illustrated in FIG. 1, a suitably sized hose clamp **86** fastens the intake tube **84** within a flange of the air filter **68**.

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In the embodiment illustrated in FIG. 1, the intake tube **84** is coupled to the air intake of the engine by way of an adapter **88**. A pair of clamps **92** ensure an airtight seal is maintained between the intake tube **84** and the air intake. The intake tube **84** preferably comprises a shape and size suitable for conducting air drawn through the air filter **68** into the air intake of the engine. As such, the intake tube **84** generally comprises an arrangement of one or more bends and one or more straight portions so as to connect the air filter **68** to the air intake of the engine. Further details about specific configurations of the intake tube **84**, including various components and fasteners for installing embodiments of the aircharger air intake **62** into various vehicles are disclosed in U.S. Patent Application, entitled "Aircharger Air Intake System And Method," filed on Mar. 8, 2017, and having application Ser. No. 15/453,496, which claims the benefit of, and priority to, U.S. Provisional Application, entitled "Aircharger Air Intake System And Method," filed on Mar. 8, 2016 and having application Ser. No. 62/305,391, the entirety of each of said applications being incorporated herein by reference.

FIG. 2 illustrates an exemplary embodiment of a control module **104** configured to create an increase in throttle responsiveness of a vehicle. The control module **104** generally is configured to instruct the ECU to increase the manifold pressure before releasing the pressure through the waste gate so as to maintain additional boost for an increased power output of the engine. The control module **104** may be comprised of one or more microprocessors that can process input signals received from a turbo inlet pressure (TIP) sensor **52** and a manifold absolute pressure (MAP) sensor **58** of the vehicle (see FIG. 1). As will be appreciated, the control module **104** may include hardware comprising electronic components on a printed circuit board (PCB), ceramic substrate or a thin laminate substrate, and include a micro controller chip (CPU). Software may be stored in the micro-controller or other chips on the PCB, such as EPROMs or flash memory, so that the CPU can be re-programmed by uploading updated code or replacing chips. The control module **104** preferably has a fixed programming, such as an internal lookup table whereby TIP and MAP sensor readings may be evaluated.

As shown in FIG. 2, the control module **104** includes a rigid enclosure **116** and an input socket **120**. The input socket **120** is configured to receive a signal connector comprising the wiring harness **108**, as discussed herein. The input socket **120** facilitates coupling the control module **104** with the TIP and MAP sensors **52**, **58** (FIG. 1) of the vehicle, as well as coupling the control module **104** with the ECU of the vehicle. Further, it is contemplated that the rigid enclosure **116** is configured to withstand the environment encountered within an engine compartment of the vehicle for the purpose of protecting the internal circuitry of the control module **104**.

FIG. 3 illustrates an exemplary embodiment of a wiring harness **108** configured to electrically couple the control module **104** with the TIP and MAP sensors **52**, **58** (FIG. 1) of the vehicle, as well as coupling the control module **104** with the ECU. The wiring harness **108** generally includes a cable **124**, a TIP sensor connector **128**, a TIP sensor harness connector **132**, a signal connector **136**, and a MAP sensor connector **140**. As will be recognized, the cable **124** includes an exterior sheath configured to protect the cable **124** from potential damage due to nearby components comprising the vehicle. The TIP sensor connector **128** is configured to be coupled directly with the TIP sensor **52** of the vehicle, while the TIP sensor harness connector **132** is configured to be

coupled with a TIP sensor connector **56** (see FIG. 1) of the wiring harness that was originally coupled with the TIP sensor **52**. The signal connector **136** is configured to be plugged into the input socket **120** of the control module **104**. The MAP sensor connector **140** is configured to be coupled with the MAP sensor **58** of the vehicle for the purpose reading the air pressure within the engine manifold. Thus, the wiring harness **108** effectively provides direct communication between the TIP and MAP sensors **52**, **58**, the control module **104**, and the ECU of the vehicle.

FIG. 4 illustrates an exemplary embodiment of a signal adjuster **112** configured to facilitate manual adjustment of the power output of the engine. The signal adjuster **112** comprises a cable **144** that extends from a controller connector **148** to a rheostat **152**. In the illustrated embodiment of FIG. 4, the cable **144** includes an exterior sheath configured to protect the cable **144** from potential damage due to nearby components comprising the vehicle. The controller connector **148** is configured to be plugged into a controller socket (not shown) comprising the wiring harness **108**. The rheostat **152** is configured to enable a practitioner, such as the driver, to manually interact with the control module **104** so as to control the power output of the vehicle. The signal adjuster **112** includes a control dial **156** configured to be coupled with the rheostat **152** to facilitate hand operation of the rheostat. It is contemplated that the practitioner mounts the rheostat **152** and the control dial **156** in an advantageous location within the passenger cabin of the vehicle, such as a dashboard, and then routes the cable **144** to the wiring harness **108**. The practitioner then plugs the controller connector **148** into the controller socket to place the rheostat **152** into electrical communication with the control module **104**. It is contemplated that, in some embodiments, the signal adjuster **112** may be omitted from the turbo-boost controlled intake system **60**, thereby providing a fully automated adjustment of engine power output of the vehicle.

FIG. 5 is a graph **160** illustrating manifold pressure readings by the ECU of the vehicle with and without the turbo-boost controlled intake system **60** illustrated in FIG. 1. The data plotted in the graph **160** are based on experimental observations before and after installation of the turbo-boost controlled intake system **60** into a test vehicle. During operation of the system **60**, the control module **104** continuously reads the MAP and TIP sensors **58**, **52** individually and compares the readings against an internal lookup table. The control module **104** then sends appropriate signals to the ECU. As indicated in graph **160**, if the measured manifold pressure falls within a range specified in the lookup table, a higher target manifold pressure is signaled to the ECU. It is contemplated, however, that the higher pressure is not to exceed factory vehicle limits. Graph **160** shows, therefore, that the turbo-boost controlled intake system **60** provides a desirable increase in turbo-boost as compared with the performance provided by the factory ECU.

FIG. 6 is a block diagram illustrating an exemplary data processing system **600** that may be used with an adjustable turbo-boost control system, such as the turbo-boost controlled intake system **60** to perform any of the processes or methods described herein. System **600** may represent a desktop, a tablet, a server, a mobile phone, a media player, a personal digital assistant (PDA), a personal communicator, a network router or hub, a wireless access point (AP) or repeater, a set-top box, or a combination thereof.

In an embodiment, illustrated in FIG. 6, system **600** includes a processor **624** and a peripheral interface **628**, also referred to as a chipset, to couple various components to the processor **624**, including a memory **632** and devices **636**-

648 by way of a bus or an interconnect. Processor **624** may represent a single processor or multiple processors with a single processor core or multiple processor cores included therein. Processor **624** may represent one or more general-purpose processors such as a microprocessor, a central processing unit (CPU), and the like. More particularly, processor **624** may be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processor **624** may also be one or more special-purpose processors such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal processor (DSP), a network processor, a graphics processor, a network processor, a communications processor, a cryptographic processor, a co-processor, an embedded processor, or any other type of logic capable of processing instructions. Processor **624** is configured to execute instructions for performing the operations and steps discussed herein.

Peripheral interface **628** may include a memory control hub (MCH) and an input output control hub (ICH). Peripheral interface **628** may include a memory controller (not shown) that communicates with a memory **632**. The peripheral interface **628** may also include a graphics interface that communicates with graphics subsystem **634**, which may include a display controller and/or a display device. The peripheral interface **628** may communicate with the graphics device **634** by way of an accelerated graphics port (AGP), a peripheral component interconnect (PCI) express bus, or any other type of interconnects.

An MCH is sometimes referred to as a Northbridge, and an ICH is sometimes referred to as a Southbridge. As used herein, the terms MCH, ICH, Northbridge and Southbridge are intended to be interpreted broadly to cover various chips that perform functions including passing interrupt signals toward a processor. In some embodiments, the MCH may be integrated with the processor **624**. In such a configuration, the peripheral interface **628** operates as an interface chip performing some functions of the MCH and ICH. Furthermore, a graphics accelerator may be integrated within the MCH or the processor **624**.

Memory **632** may include one or more volatile storage (or memory) devices, such as random access memory (RAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), static RAM (SRAM), or other types of storage devices. Memory **632** may store information including sequences of instructions that are executed by the processor **624**, or any other device. For example, executable code and/or data of a variety of operating systems, device drivers, firmware (e.g., input output basic system or BIOS), and/or applications can be loaded in memory **632** and executed by the processor **624**. An operating system can be any kind of operating systems, such as, for example, Windows® operating system from Microsoft®, Mac OS®/iOS® from Apple, Android® from Google®, Linux®, Unix®, or other real-time or embedded operating systems such as Vx Works.

Peripheral interface **628** may provide an interface to I/O devices, such as the devices **636**-**648**, including wireless transceiver(s) **636**, input device(s) **640**, audio I/O device(s) **644**, and other I/O devices **648**. Wireless transceiver **636** may be a WiFi transceiver, an infrared transceiver, a Bluetooth transceiver, a WiMax transceiver, a wireless cellular telephony transceiver, a satellite transceiver (e.g., a global positioning system (GPS) transceiver) or a combination thereof. Input device(s) **640** may include a mouse, a touch

pad, a touch sensitive screen (which may be integrated with display device 634), a pointer device such as a stylus, and/or a keyboard (e.g., physical keyboard or a virtual keyboard displayed as part of a touch sensitive screen). For example, the input device 640 may include a touch screen controller coupled with a touch screen. The touch screen and touch screen controller can, for example, detect contact and movement or break thereof using any of a plurality of touch sensitivity technologies, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with the touch screen.

Audio I/O 644 may include a speaker and/or a microphone to facilitate voice-enabled functions, such as voice recognition, voice replication, digital recording, and/or telephony functions. Other optional devices 648 may include a storage device (e.g., a hard drive, a flash memory device), universal serial bus (USB) port(s), parallel port(s), serial port(s), a printer, a network interface, a bus bridge (e.g., a PCI-PCI bridge), sensor(s) (e.g., a motion sensor, a light sensor, a proximity sensor, etc.), or a combination thereof. Optional devices 648 may further include an imaging processing subsystem (e.g., a camera), which may include an optical sensor, such as a charged coupled device (CCD) or a complementary metal-oxide semiconductor (CMOS) optical sensor, utilized to facilitate camera functions, such as recording photographs and video clips.

Note that while FIG. 6 illustrates various components of a data processing system, it is not intended to represent any particular architecture or manner of interconnecting the components; as such details are not germane to embodiments of the present disclosure. It should also be appreciated that network computers, handheld computers, mobile phones, and other data processing systems, which have fewer components or perhaps more components, may also be used with embodiments of the invention disclosed hereinabove.

Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities.

It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the above discussion, it should be appreciated that throughout the description, discussions utilizing terms such as those set forth in the claims below, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system's memories or registers or other such information storage, transmission or display devices.

The techniques shown in the figures can be implemented using code and data stored and executed on one or more electronic devices. Such electronic devices store and communicate (internally and/or with other electronic devices over a network) code and data using computer-readable

media, such as non-transitory computer-readable storage media (e.g., magnetic disks; optical disks; random access memory; read only memory; flash memory devices; phase-change memory) and transitory computer-readable transmission media (e.g., electrical, optical, acoustical or other form of propagated signals-such as carrier waves, infrared signals, digital signals).

The processes or methods depicted in the preceding figures may be performed by processing logic that comprises hardware (e.g. circuitry, dedicated logic, etc.), firmware, software (e.g., embodied on a non-transitory computer readable medium), or a combination of both. Although the processes or methods are described above in terms of some sequential operations, it should be appreciated that some of the operations described may be performed in a different order. Moreover, some operations may be performed in parallel rather than sequentially.

While the invention has been described in terms of particular variations and illustrative figures, those of ordinary skill in the art will recognize that the invention is not limited to the variations or figures described. In addition, where methods and steps described above indicate certain events occurring in certain order, those of ordinary skill in the art will recognize that the ordering of certain steps may be modified and that such modifications are in accordance with the variations of the invention. Additionally, certain of the steps may be performed concurrently in a parallel process when possible, as well as performed sequentially as described above. To the extent there are variations of the invention, which are within the spirit of the disclosure or equivalent to the inventions found in the claims, it is the intent that this patent will cover those variations as well. Therefore, the present disclosure is to be understood as not limited by the specific embodiments described herein, but only by scope of the appended claims.

What is claimed is:

1. A turbo-boost controlled intake system for providing greater control over vehicle performance, the system comprising:

- an aircharger air intake system for conducting a filtered airstream to an air intake of an engine;
- a control module for interpreting signals and communicating with an electronic control unit of the vehicle;
- a wiring harness for communicating signals among one or more sensors and the control module;
- a turbo inlet pressure sensor connector to be coupled directly with a turbo inlet pressure sensor of the vehicle; and
- a turbo inlet pressure sensor harness connector to be coupled with a vehicle wiring harness that was originally coupled with the turbo inlet pressure sensor.

2. The system of claim 1, wherein the wiring harness comprises: a cable that electrically couples the control module with the turbo inlet pressure sensor, the manifold absolute pressure sensor, and the electronic control unit of the vehicle; a signal connector for plugging into an input socket of the control module; and a manifold absolute pressure sensor connector configured to be coupled with a manifold absolute pressure sensor of the vehicle.

3. The system of claim 1, further comprising a signal adjuster configured to enable manual adjustment of the power output of the engine.

4. The system of claim 3, wherein the signal adjuster includes a rheostat for enabling the manual adjustment.

5. The system of claim 4, wherein the signal adjuster includes a control dial to facilitate hand operation of the rheostat.

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6. The system of claim 4, wherein the signal adjuster includes a cable extending from the rheostat to a controller connector to be plugged into a controller socket comprising the wiring harness.

7. The system of claim 1, wherein the control module includes an input socket that receives a signal connector comprising the wiring harness.

8. The system of claim 7, wherein the signal connector couples the control module with the turbo inlet pressure sensor, the manifold absolute pressure sensor, and the electronic control unit of the vehicle.

9. The system of claim 8, wherein the control module is configured to process input signals received from the turbo inlet pressure sensor and the manifold absolute pressure sensor.

10. A method for a turbo-boost controlled intake system to provide greater control over vehicle performance, the method comprising:

configuring an aircharger air intake system to conduct a filtered airstream to an air intake of an engine;

configuring a control module to interpret signals and communicate with an electronic control unit of the vehicle;

configuring a wiring harness to communicate signals among one or more sensors and the control module;

configuring a turbo inlet pressure sensor connector to be coupled directly with a turbo inlet pressure sensor of the vehicle; and

configuring a turbo inlet pressure sensor harness connector to be coupled with a vehicle wiring harness that was originally coupled with the turbo inlet pressure sensor.

11. The method of claim 10, wherein configuring the wiring harness comprises: configuring a cable to electrically

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couple the control module with the turbo inlet pressure sensor, the manifold absolute pressure sensor, and the electronic control unit of the vehicle; configuring a signal connector to be plugged into an input socket of the control module; and configuring a manifold absolute pressure sensor connector to be coupled with a manifold absolute pressure sensor of the vehicle.

12. The method of claim 10, further comprising configuring a signal adjuster to enable manual adjustment of the power output of the engine.

13. The method of claim 12, wherein configuring the signal adjuster includes configuring a rheostat for enabling the manual adjustment.

14. The method of claim 13, wherein configuring the signal adjuster includes configuring a control dial to facilitate hand operation of the rheostat.

15. The method of claim 13, wherein configuring the signal adjuster includes routing a cable from the rheostat to a controller connector to be plugged into a controller socket comprising the wiring harness.

16. The method of claim 10, wherein configuring the control module includes configuring an input socket to receive a signal connector comprising the wiring harness.

17. The method of claim 16, wherein configuring the control module includes configuring the signal connector to couple the control module with the turbo inlet pressure sensor, the manifold absolute pressure sensor, and the electronic control unit of the vehicle.

18. The method of claim 17, wherein configuring the control module includes configuring the control module to process input signals received from the turbo inlet pressure sensor and the manifold absolute pressure sensor.

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